Pauli on, 194
problems in, 194–195
uncertainty principle in, 195–196
wave functions in, 193
Bohr, Niels, 28, 31, 35, 54–58, 66–67, 74–75, 109
Boltzmann, Ludwig, 158–163, 276
Born, Max, 20, 33, 58–59, 65–66, 67
Born rule
description of, 19–20
in Many-Worlds theory, 146–148
particle locations and, 191–193
probabilities and, 130–131, 145, 167–168
as Pythagoras's theorem, 87, 142
self-locating uncertainty, 171
Bousso, Raphael, 300
brains, as coherent physical systems, 220
branch counting, 142–144
branching
atoms and, 138–139
cause of, 213–214
decision making and, 213–216
decoherence and, 119–120, 122–123, 137–138, 183, 186
description of, 157–161
as emergent worlds, 239
with four consecutive spin measurements, 134
in Many-Worlds theory, 169–172
Many-Worlds theory and, 138–140
as nonlocal process, 171–172
quantum systems and, 216
Schrödinger's equation and, 116
Bunn, Ted, 117

Cao, ChunJun (Charles), 285
categorical imperative (Kant), 210
Caves, Carlton, 198
CFT (conformal field theory), 303
choice-making. *See* free will
classical electromagnetism, 250, 269
classical mechanics
atoms (*See* atoms)
fields (*See* fields)
Hamiltonian, 64, 196, 239–240, 281–282
momentum in, 16, 70–71, 239
Newton and, 14–15
particles (*See* particles)
position in, 16, 70–71, 239
rules of, 21–22
closed universe, 287–288
cloud of probability, 19, 37
Coleman, Sidney, 129